Customer Churn Analysis for Retention Strategy

Power BI Business Analytics Portfolio Case Study

Project Overview

This project analyses customer churn for a fictional telecommunications provider, ConnectTel Australia, using a public Telco Customer Churn dataset. The dataset contains 7,043 customer records and 21 features, with Churn used as the target field for identifying customers who left the business.

The goal of the project is to help business stakeholders understand where churn is happening, which customer segments are most at risk, how much recurring revenue is exposed, and which retention actions should be prioritised.

I. Business Problem

ConnectTel Australia is experiencing a high level of customer churn, which is reducing monthly recurring revenue and increasing pressure on customer acquisition. Although the business has customer, service, contract, payment, and revenue data available, management does not have a clear view of:

- which customer segments are most likely to leave
- which churn drivers are structural versus behavioural
- which segments create the largest financial risk
- where the retention team should focus first

Without this visibility, retention efforts may be too broad, reactive, and inefficient.

II. Problem Statement

ConnectTel Australia is experiencing material customer churn that is eroding recurring revenue and reducing retention efficiency. The business lacks clear visibility into which customer groups are most likely to churn, which commercial and service factors are linked to churn, and where intervention efforts should be prioritised. This Power BI report aims to identify high-risk customer segments, quantify monthly revenue at risk, and support targeted retention strategy.

III. Project Objective

The objective of this project is to build an interactive Power BI dashboard that helps stakeholders:

- Monitor overall churn performance
- Identify the customer groups most likely to leave

- Understand key churn drivers across contract, tenure, payment method, internet service, and support services
- Quantify monthly revenue at risk
- Prioritise retention actions based on churn risk and financial impact

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IV. Tools and Techniques Used

Tools

- Power BI
- Power Query
- DAX
- Data modelling
- Data visualisation
- Business analytics storytelling

Key Power BI techniques

- KPI cards
- Slicers
- Calculated columns
- DAX measures
- Segmentation tables
- Matrix visuals
- Scatter plot risk mapping
- Revenue-risk analysis
- Executive dashboard design

V. Dashboard Structure

The final dashboard contains three report pages.

Page 1: Executive Overview

This page provides a high-level summary of churn performance, customer volume, active customers, monthly recurring revenue, revenue at risk, and the top at-risk customer segments.

It answers:

- What is the overall churn rate?

- How many customers have churned?
- How much revenue is at risk?
- Which contract type has the most churned customers?
- Which internet service has the highest churn rate?
- Which payment method is most associated with churn?
- Which tenure bands have the highest churn risk?
- Which segments should management prioritise first?

Page 2: Customer Segmentation Deep Dive

This page investigates churn patterns across customer profile, household characteristics, payment behaviour, internet service, support services, and monthly charge bands.

It answers:

- Do senior customers churn more than non-senior customers?
- Do customers without partners or dependents churn more?
- Which contract and payment combinations are highest risk?
- Does online security reduce churn?
- Does tech support reduce churn?
- Are higher-paying customers also more likely to churn?

Page 3: Churn Drivers & Revenue Risk

This page focuses on financial exposure and commercial impact. It identifies which segments, contract types, payment methods, and services contribute the most monthly revenue at risk.

It answers:

- Which contract type contributes the most revenue at risk?
- Which payment method creates the largest financial exposure?
- Which internet service contributes the most revenue risk?
- Which tenure bands are most financially exposed?
- Which customer segments combine high churn rate and high revenue risk?

VI. Key Business Questions and Answers

6.1 Where is churn happening most?

By churn volume, churn is happening most among month-to-month customers, with 1,655 churned customers, representing approximately 88% of all churned customers.

This indicates that low-commitment contract structures are the strongest concentration point of customer loss.

From a churn-rate perspective, the highest-risk lifecycle group is 0–6 months tenure, with a churn rate of 52.94%. Churn is also high among fiber optic customers, who have a churn rate of 41.89%.

Business interpretation:
Churn is concentrated in customers who have lower contract commitment and are earlier in their customer lifecycle. This suggests that onboarding, early engagement, and contract structure are critical retention levers.

6.2 Which customer groups are most likely to leave?

The customer groups most likely to leave are:

- month-to-month customers
- fiber optic customers
- electronic check customers
- customers in the first 6–24 months
- customers without tech support
- customers without online security
- senior customers
- customers without partners
- customers without dependents

The highest-risk segment in the dashboard is:

0–6 months | MTM | Fiber optic

This segment has:

- 74.15% churn rate
- 459 churned customers
- $37,150.15 monthly revenue at risk

Important analytical note:
The dashboard shows that these factors are individually associated with higher churn. It does not prove that every factor combined into one customer profile is the single highest-risk group unless a specific combined segment is created and tested.

Business interpretation:
The most churn-prone customers are those with low commitment, high service dependency, limited support adoption, and early customer tenure.

6.3 Which churn drivers are behavioural versus structural?

The dashboard shows both structural and behavioural churn drivers.

Structural churn drivers

These are linked to the customer's contract, lifecycle stage, or service type:

- Contract type
  Month-to-month customers churn much more than one-year or two-year customers.
- Tenure
  Churn is highest in the first 6 months and decreases as tenure increases.
- Internet service type
  Fiber optic customers have much higher churn than DSL or no-internet customers.
- Customer profile
  Senior status, partner status, and dependents status are associated with different churn levels.

Behavioural churn drivers

These are more directly actionable through business intervention:

- Payment method
  Electronic check customers have the highest churn rate at 45.29%.
- Tech support adoption
  Customers without tech support churn at 41.64%, compared with 15.17% for customers with tech support.
- Online security adoption
  Fiber optic customers without online security churn at 49.36%, compared with 21.81% for fiber customers with online security.
- Monthly charge level
  Customers in higher monthly charge bands show elevated churn rates, especially the $60–$89 and $90+ bands.

Business interpretation:
Structural drivers help identify where churn is concentrated, while behavioural drivers show where the business can intervene more directly.

6.4 Which churn segments matter most financially?

The most financially important churn segments are not simply those with the highest churn rate. They are the segments that combine:

- high churn rate
- high number of churned customers
- high monthly revenue at risk

The top revenue-risk segments are:

Segment	*Churn Rate*	*Churned Customers*	*Monthly Revenue at Risk*
0–6 months | MTM | Fiber optic	74.15%	459	$37,150.15
25–48 months | MTM | Fiber optic	43.38%	226	$20,762.05
13–24 months | MTM | Fiber optic	50.59%	215	$19,046.90
7–12 months | MTM | Fiber optic	61.95%	184	$16,028.15
0–6 months | MTM | DSL	49.49%	245	$10,999.65

Together, these top five segments account for approximately $103.99K in monthly revenue at risk.

At a broader level:

- Month-to-month contracts account for approximately $0.12M in monthly revenue at risk.
- Fiber optic customers account for approximately $0.11M in monthly revenue at risk.
- Electronic check customers account for approximately $84K in monthly revenue at risk.

Business interpretation:
The most financially important churn issue is concentrated in month-to-month fiber optic customers, especially those in early and mid-tenure groups.

6.5 Where should the retention team focus first?

The retention team should focus first on customer segments that combine:

- high churn rate
- high customer volume
- high revenue at risk
- actionable business drivers

The first retention priority should be:

Early-tenure month-to-month fiber optic customers

This includes:

- 0–6 months | MTM | Fiber optic
- 7–12 months | MTM | Fiber optic
- 13–24 months | MTM | Fiber optic

A secondary priority should be:

Early-tenure month-to-month DSL customers

Within these groups, the retention team should pay special attention to customers who:

- pay using electronic check
- do not have tech support
- do not have online security
- are in higher monthly charge bands

Business interpretation:
Retention should not be spread evenly across all customers. The business should focus on the segments where churn risk and financial impact overlap.

6.6 Which actions would likely protect the most recurring revenue?

The actions most likely to protect recurring revenue are those targeted at the strongest churn drivers.

Recommended action 1: Improve onboarding in the first 6–12 months

Churn is highest among customers in the 0–6 month tenure band. ConnectTel should strengthen early lifecycle engagement through welcome journeys, service check-ins, issue resolution, and proactive support.

Recommended action 2: Target month-to-month fiber customers

Month-to-month fiber customers represent the largest churn and revenue-risk concentration. These customers should receive proactive retention offers, contract-conversion incentives, and value reinforcement messaging.

Recommended action 3: Move electronic check customers to autopay

Electronic check customers have the highest churn rate and largest payment-related revenue exposure. The business should encourage migration to more stable payment methods such as credit card or bank transfer.

Recommended action 4: Promote tech support and online security bundles

Customers without tech support or online security show materially higher churn. Bundled support or security offers could improve perceived value and reduce churn risk.

Recommended action 5: Monitor top revenue-risk segments monthly

The top five revenue-risk segments represent a large share of exposed revenue. These segments should be tracked as monthly retention KPIs.

VII. Key Insights by Dashboard Page

Page 1: Executive Overview

The dashboard shows that ConnectTel has:

- 7,043 total customers

- 1,869 churned customers
- 5,174 active customers
- 26.54% churn rate
- $456.12K monthly recurring revenue
- $139.13K monthly revenue at risk

The most important finding is that churn is highly concentrated among month-to-month customers, particularly those using fiber optic service.

The top at-risk segment is:

0–6 months | MTM | Fiber optic

This segment alone contributes $37.15K in monthly revenue at risk.

Page 2: Customer Segmentation Deep Dive

The segmentation page shows that churn is higher among:

- senior customers: 41.68%
- customers without partners: 32.96%
- customers without dependents: 31.28%
- customers without tech support: 41.64%
- customers without online security, especially fiber customers: 49.36%

The matrix analysis shows that the combination of month-to-month contract and electronic check payment is especially risky, with a churn rate of 53.73%.

This suggests that customer churn is not caused by one factor alone. It is often the result of multiple risk conditions appearing together.

Page 3: Churn Drivers & Revenue Risk

The revenue-risk page shows that the largest financial exposure comes from:

- month-to-month contracts
- fiber optic customers
- electronic check payment method
- early-tenure customers

Month-to-month customers account for approximately $0.12M in monthly revenue at risk, while fiber optic customers account for approximately $0.11M.

The scatter chart confirms that the most important retention opportunities are the segments that combine high churn rate with high monthly revenue exposure.

VIII. Business Recommendations

Recommendation 1: Prioritise early-tenure month-to-month fiber customers

This group has the strongest combination of churn rate, churn volume, and revenue risk. ConnectTel should create a targeted retention playbook for these customers, including onboarding support, service check-ins, and personalised offers.

Recommendation 2: Introduce contract-conversion incentives

Since month-to-month customers drive most churn, the business should test incentives that encourage customers to move into one-year or two-year contracts. These incentives could include discounts, bundled services, or loyalty benefits.

Recommendation 3: Reduce payment-related churn

Electronic check is the highest-risk payment method. ConnectTel should run a campaign encouraging electronic-check customers to switch to automatic payment methods.

Recommendation 4: Bundle tech support and online security

Customers without tech support or online security have much higher churn. The business should test whether bundling these services improves perceived value and reduces churn.

Recommendation 5: Strengthen early customer lifecycle management

Because churn is highest in the first 6 months, retention should begin immediately after acquisition. The business should monitor new customers closely and proactively resolve service, billing, and setup issues.

Recommendation 6: Build a monthly retention monitoring process

The business should track the top revenue-risk segments monthly and monitor whether retention actions reduce churn, improve contract stability, and protect recurring revenue.

IX. Business Value Delivered

This project provides value by helping ConnectTel move from general churn reporting to targeted retention decision-making.

The dashboard enables stakeholders to:

- identify where churn is concentrated
- compare churn across customer groups
- separate high-risk segments from high-value risk segments
- quantify monthly revenue exposure
- prioritise retention actions based on business impact

Instead of treating all churned customers equally, the business can now focus on the segments that matter most.

X. Limitations

This analysis is based on a sample churn dataset and should be interpreted as a portfolio business case rather than a live operational report.

Key limitations include:

- The dataset does not include customer interaction history, complaints, call centre records, or satisfaction scores.
- The analysis identifies associations, not proven causation.
- Revenue at risk is estimated using monthly charges.
- The dashboard does not include acquisition cost or customer lifetime value.
- Retention recommendations should be validated through business experiments or A/B testing before full rollout.

XI. Next Steps

The next stage of the project would be to extend the dashboard from descriptive analytics into retention planning.

Recommended next steps:

1. Create a retention priority scoring model.
2. Add customer lifetime value or profitability if available.
3. Add customer complaint or support ticket data.
4. Build a churn prediction model using Python or Power BI.
5. Test retention campaigns for high-risk segments.
6. Monitor churn reduction month by month after intervention.